

PungentFunk Utilities - Architecture and Design Principles Bible

Stable architecture/design doctrine separated from volatile script snapshots

Separated from the volatile snapshot/index. This document records stable architecture and design principles for PungentFunk Utilities. Update this bible when the philosophy, package boundaries, UI doctrine, or interoperability model changes. Update the snapshot audit separately whenever scripts change.

1. Scope and North Star

PungentFunk Utilities should be treated as a family of Unity editor/runtime utility assets, not as a single monolithic toolbox. Each lab or suite should be independently releasable, while the larger bundle should expose a cohesive control panel, shared visual language, and optional cross-lab enhancements.

The core promise is simple: automate tedious game-development workflows without adding new tedium. Every tool should be understandable at a glance, safe to try, and powerful enough to scale through presets, profiles, contextual help, and visual feedback.

Release model

- **Core package:** shared registry, lab launcher, theme system, preferences, common editor UX helpers, optional integration contracts, and documentation patterns.
- **Standalone lab packages:** independently useful products such as Colour Lab, Audio Lab, Asset Placement Lab, Texture Lab, Scene Workflow Lab, Asset Preview and Export Lab, Project Audit and Authoring Lab, Content Generation Lab, UI and Feedback Lab, and later Environment/Settings/Action labs.
- **Bundle package:** installs all labs and unlocks cross-lab linking, shared dashboards, richer context menus, and handoff workflows between tools.
- **Project adapters:** thin game-specific wrappers that pre-fill generic utility settings, preserve old menu paths, or translate project-specific data into generic profiles without polluting generic packages.

Definition of a PungentFunk utility

- It solves a broadly reusable Unity workflow problem.
- It uses generic names, generic data models, and user-configurable presets instead of hardcoded project assumptions.
- It works in a clean project, or clearly declares optional dependencies that can be absent without compile/runtime errors.
- It has preview/dry-run behavior before destructive changes, and records Undo where scene or asset changes are made.
- It uses the shared visual language and persistence helpers unless there is a deliberate, documented reason not to.

2. Product Architecture

Layer	Purpose	Rules
PungentFunk Utilities Core	Registry, launcher/control panel, lab menu routing, shared theme, preferences, shared widgets, package status metadata.	Every lab may depend on Core. Core must not depend on any lab. Keep Core small, stable, and boring.
Runtime model assemblies	ScriptableObjects, MonoBehaviours, data models, adapters, and runtime services used outside the Unity Editor.	No UnityEditor references. No editor-only menu logic. Use stable IDs and version fields.
Editor lab assemblies	Editor windows, inspectors, scanners, generators, validators, and authoring panels.	May depend on UnityEditor and Core. Scan/apply operations must be explicit, cancelable where expensive, and cached.
Optional bridges/adapters	Code that connects two labs or a lab to third-party packages.	Must be removable. Use version defines, reflection, asmdef constraints, or isolated bridge packages.
Project adapters	Game-specific wrappers and presets outside the generic asset.	Thin only. They may reference game classes; generic packages must never compile against them.

Dependency direction

- Allowed: Lab -> Core, Editor -> Runtime, Project Adapter -> Generic Lab, Bridge -> Lab A + Lab B.
- Discouraged: Lab A directly compiles against Lab B unless Lab B is a declared required dependency for that product.
- Forbidden: Core -> Lab, Runtime -> Editor, Generic Lab -> game-specific project class, and cyclic lab dependencies.
- Preferred optional link: detect availability by registry ID, type lookup, define symbol, or optional bridge package; show a helpful disabled state when absent.

Labs should own their domain models and workflows. Shared abstractions should move to Core only when at least two labs truly need them and they are stable enough to support as public API. Avoid pre-emptively centralising immature helpers.

3. Editor UI and UX Principles

The suite should have a cohesive visual identity, but not a rigid one. Consistency should come from shared theme colours, typography rules, spacing scale, button/status/box styles, icons, interaction conventions, safety patterns, registry metadata, and lab navigation. Consistency should not mean every workflow is forced into the same vertical inspector layout.

Responsive layout rules

- The window must remain usable at small sizes. Do not assume a wide monitor.
- Prefer vertical scrollbars over horizontal scrollbars. Avoid horizontal scrolling unless the data is inherently matrix-like, timeline-like, graph-like, or a deliberate canvas workspace.
- Use wrapping rows, compact cards, collapsible sections, and column stacking before resorting to horizontal scrolling.
- All important shared borders between subpanels should be draggable where panel proportions matter; persist dimensions through `UtilityWindowPrefs`.
- Avoid dead space. If a row or card has large blank interior gaps, shrink padding, wrap controls, or collapse secondary metadata.
- Controls should not jump around during refresh. Important actions should not be buried below a large scroll region.

Visual information design

- Use `UtilityWindowTheme` for headers, panels, section titles, count/status pills, tinted buttons, warning/info/help boxes, splitter handles, and shared text styles.
- Use colour intentionally for status, category, severity, selection, or lab identity. Avoid unrelated colour noise.
- Use symbols, icons, tags, small badges, lock markers, warning markers, and count chips for repeated states.
- Use diagrams, mini graphs, swatch grids, matrices, timelines, or heatmaps only when they clarify relationships faster than text.
- Use concise text in the main panel; put long explanations in foldouts, documentation popups, or help panels.

4. Window and Panel Layout Design Principles

Each window and panel should be designed deliberately. A utility layout is part of its functionality, not just presentation. The correct layout must be chosen based on workflow, information density, user intent, frequency of use, preview needs, risk level, and comparison needs.

Layouts should be purpose-built

- Identify whether the user is configuring settings, scanning assets, comparing results, editing a selected item, previewing visual output, running a batch operation, iterating on generated content, monitoring runtime/debug state, navigating modules, authoring data assets, or applying scene/asset changes.
- The layout should support that workflow directly. Avoid defaulting every tool to a long vertical inspector-style panel.
- Inspector-like layouts are useful for simple configuration, but not always appropriate for comparison, preview, generation, search, graph-like relationships, or batch review.

Template	Best for	Design notes
Simple vertical stack	Small settings, single-purpose utilities, lightweight helpers.	Header -> configuration -> actions -> output/status/help. Use for linear workflows only.

Template	Best for	Design notes
Two-column layout	Source/result, configuration + preview, selection + detail.	Left controls/filter/list; right results/preview/details. Use persistent draggable splitters.
Three-panel layout	Advanced authoring, asset browsers, debugging, lab hubs.	Left navigation/source, centre workspace, right details/actions. Collapse/stack at narrow sizes.
Grid/card layout	Asset sets, swatches, prefab previews, icons, generated names, tool launchers.	Cards wrap vertically; use badges, colour chips, icons, and compact metadata.
Table/matrix layout	Coverage tools, validation, dependency maps, compatibility checks, batch reports.	Sorting/filtering/grouping required. Horizontal scroll is acceptable only when column relationships matter.
Canvas/lab workspace	Generation tools, asset placement, node-like/spatial workflows, preview-first creative tools.	Central work area, side panels, contextual toolbar, overlays, mode controls, collapsible advanced settings.
Wizard/step flow	Setup, migration, export, risky batch operations, multi-stage generation.	Show current step, required inputs, validation, preview/dry-run, and final apply/export.
Dashboard/overview	Control panel, lab home, health summaries, scan summaries.	Summary cards, status pills, warnings, search, recent/pinned tools, quick actions. Orientation and triage, not dense control.
Contextual mini panel/inline toolbar	Component tools, terrain helpers, scene-selection tools, asset-specific quick actions.	Compact controls embedded near the relevant Unity context with an Open Full Tool handoff.

Layout selection principles

- **Task frequency:** common actions should be closer and more visible than rare actions.
- **Risk level:** destructive/project-wide actions need preview, confirmation, and visual separation.
- **Information density:** dense information should be grouped, filtered, summarised, and progressively disclosed.
- **Comparison needs:** if the user must compare multiple items, avoid hiding them behind separate foldouts.
- **Preview needs:** if visual output matters, give preview space priority.
- **Selection flow:** if the user selects an item and edits details, use list/detail or grid/detail patterns.
- **Window size:** layouts must degrade gracefully at smaller widths.
- **Scanning speed:** use alignment, spacing, icons, and colour to help the user understand state quickly.
- **Cognitive load:** prefer fewer visible decisions at once, with advanced options collapsed or contextual.
- **Creative flow:** generation/design tools should support iteration without burying regenerate, lock, save, preview, or apply actions.

External design disciplines to draw from

- Professional software design: hierarchy, predictable controls, persistent layout, familiar interaction models.
- Web application design: responsive layouts, cards, dashboards, tabs, sidebars, search/filter patterns, progressive disclosure.
- Graphic design: alignment, spacing, contrast, rhythm, grouping, visual hierarchy, restrained colour usage.
- Infographic design: icons, legends, status symbols, compact summaries, visual explanation of complex relationships.
- Poster/editorial design: strong headers, readable sections, clear calls to action, intentional whitespace.
- Data visualization: matrices, charts, graphs, heatmaps, legends, tooltips, meaningful colour scales.
- Creative software: palettes, inspectors, canvases, toolbars, contextual controls, preview-first workflows.
- Accessibility design: keyboard access, readable contrast, non-colour status indicators, scalable text, clear focus states.

Layout anti-patterns to avoid

- Long undifferentiated walls of fields; excessive nested foldouts; large empty gaps caused by rigid sizing.
- Horizontal scrolling except for justified tables/matrices/canvases; controls that jump around during refresh.
- Important actions hidden below large scroll regions; duplicate buttons doing the same thing.
- Dense text where an icon, swatch, badge, or graph would be clearer.
- Tool windows that assume a large monitor; unrelated workflows mixed in one panel without grouping.
- Making every utility look identical at the cost of usability.

5. Progressive Disclosure, Assistance, and Accessibility

Tools should be understandable at a glance, but deep enough for advanced users. Show the core workflow first, keep advanced options collapsed by default, explain advanced options contextually, and never hide essential actions in advanced sections.

Tooltips and contextual help

- Every meaningful field should have a tooltip or help affordance.
- Tooltips should explain what the field controls, when to change it, how it differs from related options, and whether the value is destructive, experimental, expensive, or preview-only.
- Enum fields should provide contextual explanation for the currently selected option. Complex enums may need a small description box under the field.
- Error states should say what failed, why it likely failed, and what the user can do next.

Visual assistance and accessibility

- Use colour chips, icons, graphs, diagrams, mini previews, status pills, and legends where they reduce cognitive load.
- Do not rely on colour alone. Pair colour with icons, labels, patterns, shapes, or tooltips.
- Maintain contrast, focus visibility, keyboard accessibility where applicable, and readable scalable text.
- Prefer plain-language labels for common operations; use technical terms where they are domain-standard and useful.
- Expose high-risk/destructive actions with obvious visual separation, confirmation, and preview/dry-run patterns.

6. Editor Access and Integration Principles

Menu items are the baseline access method, not the only access method. Tools should appear where the user naturally needs them, while avoiding duplicate clutter.

Access method	Principle
Organised menu paths	Use clean Tools/PungentFunk Utilities/... or Tools/Utilities/... roots. Avoid duplicate entries, old project-specific names, and deep nesting without reason. Legacy aliases should open new generic implementations only.
Utility Control Panel / Lab Hub	Every significant utility should register with the central registry. The hub should support search, lab/module/status filtering, recent/pinned tools, lab descriptions, missing optional dependencies, and context-aware launch actions.
Unity Editor toolbar	Expose compact quick-launch buttons, active lab shortcuts, debug/preview toggles, and status indicators only for frequent workflows. Do not duplicate the full menu tree.
Scene View overlay menu	Scene-related tools should use proper overlay integration where useful: placement, surface alignment, scene navigation, gizmo browsing, debug visual toggles, path authoring, terrain/surface helpers.
Right-click context menus	Use concise context actions for relevant scene objects, components, assets, folders, materials, textures, prefabs, ScriptableObjects, terrains, audio assets, palettes, placement groups, and generated markers.
Inspector-embedded toolbars	Use compact toolbar rows, validation/status panels, quick actions, mini previews, dependency messages, and Open in Lab buttons in related components/assets.
Cross-utility embedding	Expose related tools inside other PungentFunk labs through service interfaces, reflection-safe adapters, package detection, optional asmdef references, or bridge packages.
Shortcuts and command palette	Support shortcuts for frequent actions such as opening the control panel, toggling debug tools, scanning selection, regenerating output, validating assets, or applying selected previews.
Selection/context-aware launching	When opened from selected terrain, prefab, palette, audio profile, scene object, or placement group, preload the relevant context and fall back gracefully when context is absent.
Recent tools and quick actions	Support recently opened tools, pinned tools, favourite labs, last active module per lab, resume last session, and related-tool handoffs through shared preferences.

7. Registry, Navigation, and Lab Hubs

The registry is the package-level discovery contract. Every utility should have an ID, display name, lab, module, category, sort order, status, description, tags, menu path, capability flags, and optional related utility IDs. Lab hubs should organise tools without removing direct window access.

Field	Purpose
Id	Stable machine-readable identifier. Use kebab-case. Never reuse for a different tool.
DisplayName	User-facing name. Generic, clear, and asset-store friendly.

Field	Purpose
Lab / Module / Category	Lab is product area; module is subdomain; category is menu/search taxonomy.
Description	One-sentence promise: what the tool does and when to use it.
PackageStatus	Stable, Experimental, In Progress, Deprecated, or Project Adapter.
RelatedUtilityIds	Complementary tool links. Must not imply a hard dependency unless explicitly declared.
Capability flags	Supports selection, context menu, scene overlay, batch actions, or lab hub behaviour.

Menu policy

- Primary editor menus should live under Tools/PungentFunk Utilities/... or Tools/Utilities/... consistently for the chosen release line.
- Asset context menus should use Assets/PungentFunk Utilities/... .
- GameObject context menus should use GameObject/PungentFunk Utilities/... .
- CreateAssetMenu paths should use one standard root: PungentFunk Utilities//.
- Legacy aliases are allowed only as compatibility routes into the new generic tool and should be marked internally.

8. Optional Dependency and Cross-Utility Integration Principles

PungentFunk Utilities should be modular enough for each lab or suite to ship independently, while also becoming more powerful when multiple labs are installed together. Cross-lab links should enhance a tool when another lab is installed, but must not make the base lab fragile.

Dependency levels

- **Required dependencies:** minimal dependencies that must exist for the tool to compile and function, such as UnityEngine, UnityEditor for editor-only windows, and Core if the package includes shared registry/theme infrastructure.
- **Optional integrations:** features that activate when another PungentFunk lab or compatible package is installed. These must not produce compile errors when absent.
- **Suggested companions:** utilities that enhance a workflow but are not required. Missing companion messages should be helpful, calm, and non-blocking.

Optional integration techniques

- Assembly definition optional references, version defines, reflection-safe discovery, interface packages, adapter classes in separate integration folders, registry-based service lookup, ScriptableObject loose references, conditional compilation symbols, and soft dependency metadata in descriptors.

Lab / utility	Required base	Optional enhancement points
Utility Core	None	Discovers installed labs; may show status and missing optional bridge packages.
Debug Lab	Core	Scene Workflow for gizmo sources/navigation; Project Audit for debug coverage reports; UI Feedback for notifications.
Asset Placement Lab	Core; placement runtime models	Colour Lab for palette variants; Texture Lab for masks; Scene Workflow for navigation/handles; Project Audit for validation/repair; Audio Lab for surface metadata.
Scene Workflow Lab	Core	Debug Lab for gizmo diagnostics; Asset Placement for path/placement handoff.
Colour Lab	Core; colour runtime models	Texture Lab for palette-to-texture masks; Asset Placement for material variation; UI Feedback for theme previews.
Texture Lab	Core; texture generation models	Colour Lab for palette sources; Asset Preview/Export for previews and packaging.
Audio Lab	Core; audio runtime models	Project Audit for coverage validation; Asset Placement/Scene Workflow for contextual surface tagging; Settings Lab for mixer/runtime config.
Project Audit and Authoring Lab	Core	Inspects other installed labs through optional scan modules; should not own their runtime models.
UI and Feedback Lab	Core	Input System/TMP optional or guarded; runtime presenters remain layout-agnostic.

Dependency tracking

- For each lab/utility, track required PungentFunk systems, optional integrations, external Unity package dependencies, Unity version/package requirements, runtime/editor-only status, standalone/bundle suitability, provided services, and consumed services.

- The dependency map should inform Asset Store package splitting, assembly definition boundaries, menu organisation, documentation, QA testing, and future architecture decisions.

9. Runtime, Editor, Data, and Serialization Rules

Runtime package rules

- Runtime scripts must compile in a player build without UnityEditor.
- Runtime ScriptableObjects should contain data and pure runtime logic only.
- Do not store editor-window state in runtime assets unless it is meaningful to gameplay/runtime use.
- Serialized fields should have tooltips and sane defaults.
- Use stable string IDs or GUID-like keys for anything referenced across assets, adapters, or package boundaries.

Editor package rules

- Editor windows should use SerializedObject and SerializedProperty when editing assets with existing references.
- Use Undo.RecordObject / Undo.RegisterCreatedObjectUndo for scene-object changes.
- Use AssetDatabase and EditorUtility.SetDirty safely for asset creation and mutation.
- Prompt before opening/changing scenes or applying destructive project-wide changes.
- Prefer explicit Scan, Preview, Apply Selected, Apply All flows for batch operations.

Migration and compatibility

- Avoid destructive runtime class renames for components already used in scenes or prefabs.
- When replacing a project-specific tool, prefer a new generic base plus legacy wrapper/alias rather than two parallel implementations.
- Migration installers should be temporary unless they provide durable package value.
- Class wrappers and menu aliases should preserve old workflows while directing users into the new generic implementation.

10. Performance and Safety Standards

Performance

- Do not scan all scenes/assets on every OnGUI repaint.
- Cache scan results and rebuild only when settings change or the user requests refresh.
- Use EditorApplication.update queues for long-running preview/icon/export work.
- Use cancelable progress bars for expensive scans.
- Avoid Thread.Sleep in editor tools.
- Avoid SceneView.RepaintAll unless broad repaint is truly necessary. Prefer the active window or active SceneView.

Safety

- Dry-run or preview first for destructive and batch operations.
- Make dangerous toggles visually obvious and default them to safe behaviour.
- Do not overwrite existing user data unless explicitly enabled.
- Do not modify shared materials or prefab assets by default without a warning.
- When applying results, show a summary of changed, ignored, and failed items.

11. Lab Taxonomy

Lab	Product role	Primary ownership
Utility Core	Shared shell and UX infrastructure.	Registry, control panel, lab menus, tray, theme, preferences, editor performance helpers, status vocabulary, shared scan models.
Debug Lab	Runtime/editor debug control and signal observation.	Debug channels, router, signal relay, reflected fields/actions, scheduled debug calls, router panel.
Asset Placement Lab	Reusable scene asset placement and procedural dressing.	Asset sets, rules, scatter/grid/socket/group workflows, placement results, scene application.
Scene Workflow Lab	Scene authoring speed tools.	Scene navigation, gizmo browsing/sources, surface alignment, path authoring helpers.
Colour Lab	Palette creation, analysis, accessibility, and application.	Palette assets, swatches, harmony, contrast, deficiency preview, palette generation and material/UI application.
Texture Lab	Procedural texture/mask generation and array baking.	Texture settings, procedural generator, texture-array sources, bake validation.
Audio Lab	Surface/contact audio data and coverage validation.	Audio materials, clip sets, interaction profiles/matrices, terrain mappings, contact router/classifier, coverage scanners.
Asset Preview and Export Lab	Preview/render/export support for reusable assets.	Prefab icons, prefab extraction, font previews.
Project Audit and Authoring Lab	Generic authoring diagnostics and batch tools.	Reference assignment, terrain usage, tuning copy, rename, coverage matrix, token validation, tooltip notes.
Content Generation Lab	Reusable text/name generation.	Name lists, MadLib lists, name-generation window.
UI and Feedback Lab	Reusable input prompt and UI feedback seed tools.	Input prompt icon tooling now; runtime presenters should remain optional and layout-agnostic.

12. Package Hardening Rules

Namespaces

- Use package namespaces before release. Recommended roots include PungentFunk.Utilities, PungentFunk.Utilities.Runtime, PungentFunk.Utilities.Editor, PungentFunk.Utilities.Editor.Core, PungentFunk.Utilities.Editor.Theme, PungentFunk.Utilities.Placement, PungentFunk.Utilities.Audio, PungentFunk.Utilities.Colour, PungentFunk.Utilities.Generation, PungentFunk.Utilities.SceneTools, and PungentFunk.Utilities.UI.
- Do not destructively rename existing serialized runtime components unless a compatibility wrapper is provided.

Assembly definitions

- Create separate runtime and editor asmdefs for each releasable lab or Core + all labs if shipping as a single bundle first.
- Editor asmdefs reference runtime asmdefs. Runtime asmdefs never reference editor asmdefs.
- Optional bridges get their own asmdefs and dependencies, e.g. PungentFunk.Utilities.Placement.ColourBridge.
- Third-party dependencies such as TextMeshPro should be declared, optionalized, or isolated behind reflection/bridge code.

Status labels and documentation

- Use Stable, Experimental, In Progress, Deprecated, and Project Adapter consistently in descriptors, badges, and documentation.
- Each lab needs an overview, setup checklist, core concepts, recipes, troubleshooting, optional integration notes, and API/adapters page.
- Every editor window should have a Help/Notes foldout summarising what the tool changes and what it never changes.
- Every destructive or batch workflow should document its Undo support and limits.

13. Utility Definition of Done

- Registered in PungentUtilityRegistry or a consistent per-lab registrar with complete metadata.
- Primary menu path and optional context menus are cleanly organised.
- Uses UtilityWindowTheme and UtilityWindowPrefs for visual style and persistent panel state.
- Responsive at small window sizes; no avoidable horizontal scrollbar; draggable splitters where panels compete for space.

- All fields have tooltips; enum choices have contextual descriptions; complex operations have inline help.
- Runtime code compiles without UnityEditor; editor code lives in Editor folders or editor asmdefs.
- No compile-time dependency on game-specific classes.
- Optional integrations degrade gracefully when missing.
- Scan-heavy tools cache results and refresh explicitly.
- Batch/destructive operations provide preview/dry-run and Undo, or a clear warning where Undo is impossible.
- Tested in a clean project, with no optional labs installed, and with the full bundle installed.

14. Closing Doctrine

PungentFunk Utilities should remain modular in code, cohesive in user experience, and conservative in dependencies. The suite can feel interconnected without becoming tightly coupled. Core provides discovery and shared UX. Labs provide focused capability. Bridges provide optional collaboration. Project adapters preserve game-specific workflows without contaminating the generic assets.

When in doubt, choose the smaller generic utility, the thinner adapter, the safer apply flow, the clearer layout, and the more explicit dependency boundary.